

2.2. mase $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ pos def a $\langle u, v \rangle = v^T A u$

$v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ a cheme aby

$$\begin{pmatrix} v_1 & v_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = (a_{11}v_1 + a_{21}v_2)u_1 + (a_{12}v_1 + a_{22}v_2)u_2$$

vezmeme kanonicku bazi z tych vektoru $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ a $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

symetrie pro kolne k standardnimu skalarnimu soucinu a staci pro α dostaneme kdy $\begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$ a $\begin{pmatrix} -\sin \alpha \\ \cos \alpha \end{pmatrix}$

cheme aby pro napek α $\begin{pmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} -\sin \alpha \\ \cos \alpha \end{pmatrix} = 0$

$$(\cos \alpha a_{11} + \sin \alpha a_{12} \quad \cos \alpha a_{21} + \sin \alpha a_{22}) \begin{pmatrix} -\sin \alpha \\ \cos \alpha \end{pmatrix} = 0$$

$$\cos \alpha - \sin \alpha \cos \alpha a_{11} + \sin \alpha a_{12} \cos \alpha + \cos^2 \alpha a_{21} + \sin \alpha \cos \alpha a_{22} = 0$$

$$\sin \alpha \cos \alpha (a_{22} - a_{11}) + a_{12} (\cos^2 \alpha - \sin^2 \alpha) = 0 \quad \alpha = \frac{1}{2} \arctan \frac{2a_{12}}{a_{11} - a_{22}}$$

$$\frac{\sin 2\alpha}{2} (a_{22} - a_{11}) + a_{12} (\cos 2\alpha) = 0$$

$$\frac{2a_{12}}{a_{11} - a_{22}} = \tan 2\alpha$$

je definovane protoze je roven
tedy ane je to konstanta
kaze staci o α